# KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

arxiv.org/abs/2504.03553

Enabling agents to autonomously regulate knowledge utilization through situational self-awareness

# Current Agent Training Uses "Flood Irrigation"

KnowSelf

Indiscriminate knowledge injection hurts efficiency and planning quality — agents need situational self-awareness

### Trajectory Imitation

e.g., ReAct

Blindly copies gold trajectories without considering whether the agent actually needs them

**Result:** Fragile on unexpected inputs, pattern collapse

### Trial-and-Error Refinement

e.g., Reflexion

Indiscriminately triggers self-reflection regardless of whether reflection is needed

**Result:** Unnecessary inference cost, wasted compute

### Knowledge-Augmented

e.g., KnowAgent, WKM

Always injects external knowledge, even when the agent already knows the correct action

**Result:** Wasteful knowledge retrieval, can hurt performance

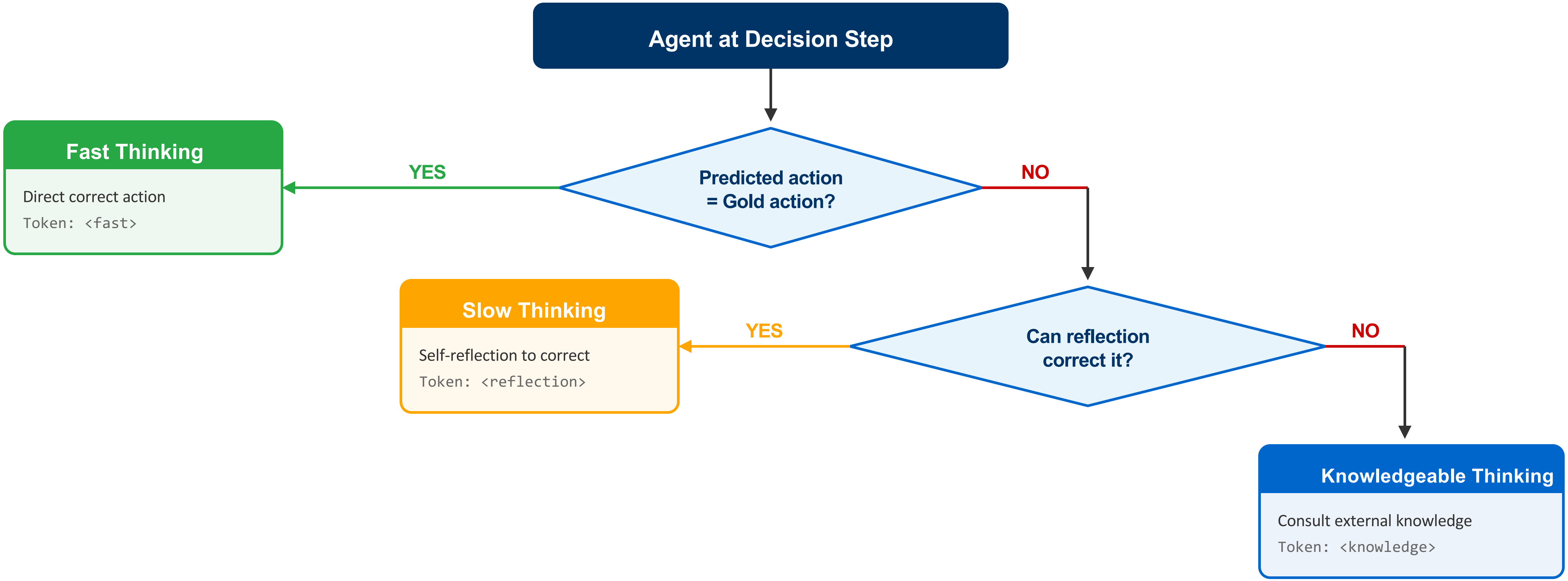
### The Human Advantage

Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to reflect, or must seek external help. KnowSelf brings this capability to LLM agents.

# Three Situational Thinking Modes

KnowSelf

KnowSelf classifies each decision step by the agent's ability level — just like human metacognition




Fast = Act directly | Slow = Reflect first | Knowledgeable = Seek help

*Decision criterion: agent's ability at each step*

# KnowSelf Framework: Data → Training → Inference



A three-stage pipeline from heuristic labeling to self-aware deployment

## STAGE 1 Data Construction

### Self-Exploration

Agent explores environment autonomously, generating trajectories for analysis

### Heuristic Judgment

At each step, classify:

- Predicted = Gold → Fast
- Reflection fixes → Slow
- Neither → Knowledgeable

### Special Token Insertion

Mark each step with <fast>, <reflection>, or <knowledge> tokens

## STAGE 2 Two-Stage Training

### Step 1: SFT

Supervised fine-tuning teaches initial self-awareness patterns from labeled trajectories

↓

### Step 2: RPO

Reward-based Policy Optimization refines the agent's ability to generate appropriate special tokens for each situation

## STAGE 3 Self-Aware Inference

### Token-Triggered Actions

Agent generates special tokens that trigger different behaviors

`<fast>` → Direct action
No additional processing needed

`<reflection>` → Self-reflection
Agent reconsiders and corrects

`<knowledge>` → Knowledge retrieval
Query external knowledge base

# KnowSelf Correctly Handles Tasks Where Baselines Fail



Reflection and knowledge tokens enable correct decisions in real interactive tasks

## Task: "Put a hot mug in cabinet"

### OpenAI-O1 — FAIL

1. Opens the microwave
2. Incorrectly tries to take an egg from the microwave
3. Wrong action sequence — fails to complete the task

### KnowSelf (Llama-8B) — PASS

1. Generates <reflection> token
2. Reconsiders the situation: "The mug needs to be heated"
3. Uses the open microwave to heat the mug correctly

`<reflection>` token triggered self-correction

## Task: "Put a saltshaker in drawer"

### DeepSeek-R1 — FAIL

1. Reflects that the drawer is open
2. Tries to close the drawer first
3. Closing fails — incorrect prioritization of actions
4. Cannot complete the task

### KnowSelf (Llama-8B) — PASS

1. Generates <knowledge> token
2. Retrieves knowledge: "Place the object without removing unrelated items"
3. Places saltshaker in drawer

`<knowledge>` token triggered external knowledge retrieval

# ALFWorld Results: 84.33% with Only 15% Knowledge



KnowSelf (Llama-8B) outperforms GPT-4o baselines while using far less external knowledge

| Backbone | Method | Know% | Put | Clean | Heat | Cool | Examine | Put Two | All |
|---|---|---|---|---|---|---|---|---|---|
| **GPT-4o** | | | | | | | | | |
| GPT-4o | ReAct | 0% | 83.33 | 74.19 | 69.57 | 85.71 | 77.78 | 64.71 | 76.12 |
| GPT-4o | Reflexion | 0% | 100.00 | 87.10 | 73.91 | 90.48 | 83.33 | 70.59 | **85.07** |
| GPT-4o | ExpeL | 100% | 95.83 | 83.87 | 69.57 | 80.95 | 88.89 | 52.94 | 79.85 |
| **Llama-8B** | | | | | | | | | |
| Llama-8B | ReAct | 0% | 33.33 | 3.23 | 0.00 | 57.14 | 66.67 | 23.53 | 27.61 |
| Llama-8B | Reflexion | 0% | 62.96 | 22.73 | 5.88 | 64.29 | 86.36 | 50.00 | 51.49 |
| Llama-8B | ExpeL | 100% | 83.33 | 32.26 | 30.43 | 23.81 | 55.56 | 17.65 | 41.04 |
| Llama-8B | ETO | 0% | 91.67 | 70.59 | 82.61 | 61.90 | 88.89 | 64.71 | 78.36 |
| Llama-8B | KnowAgent | 100% | 87.50 | 93.55 | 65.22 | 66.67 | 61.11 | 64.71 | 75.37 |
| Llama-8B | WKM | 100% | 95.83 | 87.10 | 86.96 | 61.90 | 66.67 | 52.94 | 77.61 |
| **Llama-8B** | **KnowSelf** | **15.01%** | **91.67** | **87.10** | **91.30** | **85.71** | 77.78 | 64.71 | **84.33** |
| **Gemma-2B** | | | | | | | | | |
| Gemma-2B | ReAct | 0% | 0.00 | 9.68 | 0.00 | 4.76 | 44.44 | 0.00 | 8.96 |
| Gemma-2B | ETO | 0% | 91.67 | 83.87 | 78.26 | 52.38 | 77.78 | 29.41 | 71.64 |
| Gemma-2B | KnowAgent | 100% | 91.67 | 90.32 | 69.57 | 71.43 | 66.67 | 41.18 | 73.88 |
| Gemma-2B | WKM | 100% | 91.67 | 87.10 | 78.26 | 71.43 | 61.11 | 52.94 | 76.12 |
| **Gemma-2B** | **KnowSelf** | **26.41%** | 87.50 | **93.55** | 73.91 | 76.19 | **83.33** | 52.94 | **79.85** |

## Key Observations

- KnowSelf-Llama-8B: 84.33%
  Best overall, only 15% knowledge
- KnowSelf-Gemma-2B: 79.85%
  Matches GPT-4o ExpeL at 26% knowledge
- 100% knowledge methods underperform
  ExpeL/KnowAgent/WKM use all knowledge yet achieve lower accuracy
- GPT-4o Reflexion: 85.07%
  Highest GPT-4o baseline, no knowledge

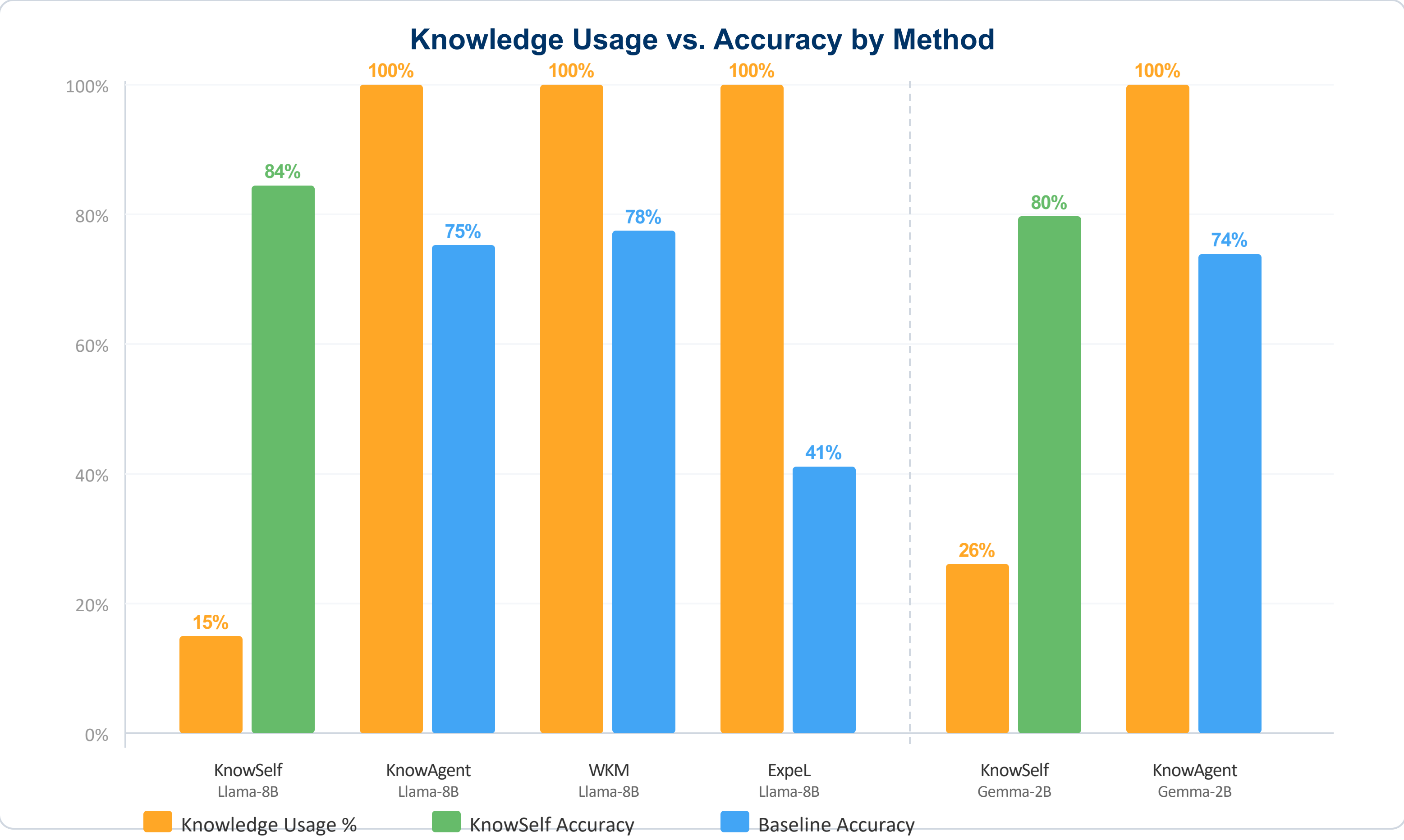
### Takeaway

Selective knowledge use outperforms blanket application across all methods

# Minimal Knowledge, Maximum Impact



KnowSelf uses 15-26% knowledge vs 100% for competitors, yet achieves the best overall accuracy




**Knowledge Used**

15%

Llama-8B KnowSelf

**vs 100% for baselines**

**Best Accuracy**

84.33%

Llama-8B KnowSelf

**Best overall score**

**Efficiency Gain**

4-6x

Less knowledge used

vs full-knowledge methods

**Scales Down**

79.85%

Gemma-2B KnowSelf

Matches GPT-4o ExpeL

### Why Less Knowledge Works Better

1. Irrelevant knowledge can mislead agents into incorrect action sequences
2. Agents that already know the correct action waste resources on retrieval
3. Selective injection preserves agent autonomy and decision quality
4. KnowSelf uses knowledge only when the agent truly cannot proceed alone

# Key Takeaways

## Self-Awareness Is the Missing Piece in Agent Planning

1. **Indiscriminate knowledge injection is wasteful and suboptimal**
   Methods using 100% knowledge (ExpeL, KnowAgent, WKM) consistently underperform selective approaches

2. **Agents can learn situational self-awareness via special tokens**
   Lightweight heuristic labeling on self-explored trajectories — no expensive human annotation required

3. **KnowSelf (Llama-8B) outperforms GPT-4o baselines**
   84.33% vs. GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) — a smaller open-source model beats GPT-4o

4. **Only 15-26% knowledge usage needed vs. 100% for competing methods**
   4-6x less knowledge consumption while achieving higher accuracy across all model scales

5. **The paradigm generalizes across model scales**
   Even Gemma-2B reaches 79.85% — matching GPT-4o ExpeL, showing the approach scales down effectively

**Selective Knowledge > Indiscriminate Knowledge**